

DISCOVERING HIDDEN RELATIONS AMONG TRIANGLE INVARIANTS

TRIANGLE-RELATIONS PROJECT

ABSTRACT. We describe a numerical method for discovering unknown functional relations among scalar quantities derived from a triangle, in the spirit of Euler’s relation between the circumradius, the inradius, and the incenter-to-circumcenter distance. A triangle has three degrees of freedom, so a relation among four or more derived scalars is guaranteed by dimension counting alone and is uninformative; a relation among exactly three is a genuine, generically unexpected coincidence. We detect such coincidences by training a bottleneck autoencoder on sampled triangle data and comparing its reconstruction error against a column-permuted null, and make the detector’s behavior mathematically precise. Applied to (R, r, d) , the method flags exactly this triple as dependent, with no prior knowledge of Euler’s relation.

1. DIMENSION COUNTING

A triangle, up to translation and rotation, has three degrees of freedom (e.g. its three side lengths a, b, c); write $\mathcal{M}$ for this moduli space, $\dim \mathcal{M} = 3$. Every derived scalar (area, R , r , distances between derived points, ...) is a smooth function $f : \mathcal{M} \rightarrow \mathbb{R}$, and k of them give a map $F = (f_1, \dots, f_k) : \mathcal{M} \rightarrow \mathbb{R}^k$. Call a property of F *generic* if it holds outside a closed, measure-zero exceptional set of choices of the f_i .

If $k > 3$, a generic F has, as its image, an (at most) 3-dimensional submanifold of $\mathbb{R}^k$: embedding a 3-dimensional space into a higher-dimensional one produces $k - 3$ constraint equations *for free*, regardless of which f_i were chosen. So a relation among four or more triangle invariants says nothing specific about triangle geometry.

If $k = 3$, recall that the Jacobian $DF(p)$ is the 3×3 matrix of partial derivatives $(\partial f_i / \partial x_j)$ in local coordinates on $\mathcal{M}$, and F is a *local diffeomorphism* at p (a smooth bijection near p with smooth inverse) exactly when $DF(p)$ is invertible. Generically DF has full rank 3, and (f_1, f_2, f_3) sweeps out an open 3-dimensional region of $\mathbb{R}^3$ — a solid blob, not a surface. An *exact* relation $G(f_1, f_2, f_3) = 0$ is equivalent to DF having rank ≤ 2 everywhere, collapsing the image onto a 2-dimensional surface: not generic, a genuine algebraic coincidence. Euler’s relation

$$d^2 = R^2 - 2Rr \tag{1.1}$$

(d = incenter-to-circumcenter distance) is exactly such a statement about (R, r, d) . This is the sense in which a relation among exactly three invariants — not four — is structurally new.

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2. THE AUTOENCODER DETECTOR

Given a candidate triple (f_1, f_2, f_3) , sample many random triangles and form the point cloud $X = \{(f_1, f_2, f_3)(T)\} \subset \mathbb{R}^3$. An autoencoder — an encoder $e : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ and decoder $d : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ trained to minimize $\mathbb{E}\|d(e(x)) - x\|^2$ — searches for the best rank-2 approximation to X : if X lies on a 2-surface, the encoder need only learn a local chart and the decoder its inverse, driving error to 0; if X fills an open 3-region, no map factoring through $\mathbb{R}^2$ can avoid discarding a whole dimension, so error stays bounded away from 0.

2.1. A precise statement. Let P be the law of $(f_1, f_2, f_3)(T)$ and $X = \text{supp}(P)$. By a *two-layer perceptron of width h* we mean $N(x) = W_2 \sigma(W_1 x + b_1) + b_2$, with $W_1 \in \mathbb{R}^{h \times p}$, $b_1 \in \mathbb{R}^h$, $W_2 \in \mathbb{R}^{q \times h}$, $b_2 \in \mathbb{R}^q$ trainable and $\sigma = \tanh$ applied componentwise; write $\mathcal{N}_h(\mathbb{R}^p, \mathbb{R}^q)$ for the class of all such maps. Recall g is *K-Lipschitz* if $\|g(x) - g(y)\| \leq K\|x - y\|$; since $\tanh$ is 1-Lipschitz, every two-layer perceptron is Lipschitz, with constant controlled by $\|W_1\|, \|W_2\|$. Train $e \in \mathcal{N}_h(\mathbb{R}^3, \mathbb{R}^2)$, $d \in \mathcal{N}_h(\mathbb{R}^2, \mathbb{R}^3)$ to minimize $L(e, d) = \mathbb{E}_{x \sim P} \|d(e(x)) - x\|^2$.

Proposition 2.1 (Sufficiency). *Recall a subset of $\mathbb{R}^n$ is compact iff closed and bounded (Heine–Borel). If X is compact and admits continuous $\varphi : X \rightarrow \mathbb{R}^2$, $\psi : \varphi(X) \rightarrow \mathbb{R}^3$ with $\psi(\varphi(x)) = x$ for all $x \in X$ — in particular whenever an exact relation confines X to a 2-dimensional set — then for every $\varepsilon > 0$ there is a width h and (e, d) with $L(e, d) < \varepsilon$.*

Proof sketch. By the universal approximation theorem (Cybenko 1989; Hornik 1991): for any continuous $\varphi : K \rightarrow \mathbb{R}^q$ on compact K and any $\delta > 0$, some $N \in \mathcal{N}_h(\mathbb{R}^p, \mathbb{R}^q)$ satisfies $\sup_{x \in K} \|N(x) - \varphi(x)\| < \delta$. Approximating φ on X and ψ on $\varphi(X)$ gives e, d with $\sup_{x \in X} \|d(e(x)) - x\| \rightarrow 0$, hence $L(e, d) \rightarrow 0$. $\square$

Proposition 2.2 (Necessity, at bounded capacity). *Fix $K > 0$. Recall P is absolutely continuous (w.r.t. Lebesgue measure) if it assigns zero probability to every Lebesgue-null set, i.e. has a density. If X contains an open ball B on which P is absolutely continuous with $P(B) > 0$, there is $c = c(P|_B, K) > 0$ with $L(e, d) \geq c$ for every pair of K -Lipschitz e, d — in particular for every $(e, d) \in \mathcal{N}_h(\mathbb{R}^3, \mathbb{R}^2) \times \mathcal{N}_h(\mathbb{R}^2, \mathbb{R}^3)$ whose realized weights keep the Lipschitz constant below K .*

Proof sketch. The Hausdorff dimension of $S \subset \mathbb{R}^n$ generalizes ordinary dimension to non-smooth sets (covering S by ε -balls takes $\sim \varepsilon^{-\dim S}$ of them); Lipschitz maps do not increase it. So $d(\mathbb{R}^2)$ has Hausdorff dimension ≤ 2 , hence is Lebesgue-null in $\mathbb{R}^3$, hence P -null on B ; thus $\text{dist}(x, d(\mathbb{R}^2)) > 0$ for $P|_B$ -a.e. x , and a compactness argument over K -Lipschitz maps gives $\mathbb{E}_{x \sim P|_B} \text{dist}(x, d(\mathbb{R}^2))^2 \geq c > 0$. Since $d(e(\mathbb{R}^3)) \subseteq d(\mathbb{R}^2)$, the same bound applies to $L(e, d)$. $\square$

Remark 2.3. By Brouwer’s invariance-of-domain theorem, a continuous injection between open subsets of $\mathbb{R}^n$ is an open map, so no continuous injection of $\mathbb{R}^2$ into $\mathbb{R}^3$ has open image: exact space-filling is topologically impossible regardless of capacity. What is not ruled out is *approximate* space-filling by increasingly large-Lipschitz-constant maps as $K \rightarrow \infty$, which is why Proposition 2.2 fixes K ; in practice ℓ_2 weight regularization and early stopping keep the realized Lipschitz constant controlled.

Propositions 2.1 and 2.2 say that, at any fixed, regularized capacity, reconstruction error going to 0 versus staying bounded away from it is decided entirely by whether

(f_1, f_2, f_3) satisfy an exact relation — the theoretical content behind the detection rule used throughout.

3. CALIBRATION VIA A PERMUTATION NULL

An absolute error threshold is not meaningful across triples: they differ in units and scale, and since all derived scalars are functions of the same three triangle parameters, some triples correlate mildly without an exact relation. We build a null X' by independently permuting each column of X , which preserves each f_i 's marginal while destroying the joint dependency. Writing L_{real} and $\mu_{\text{null}}, \sigma_{\text{null}}$ for the real and (repeated) null reconstruction risk, we compare via the z-score

$$z = \frac{\mu_{\text{null}} - L_{\text{real}}}{\sigma_{\text{null}}}, \quad (3.1)$$

or the ratio $L_{\text{real}}/\mu_{\text{null}}$: a large z (small ratio) is the fingerprint of an unexpected relation.

4. THE PIPELINE

- (1) Enumerate candidate triples of derived scalars.
- (2) Sample random triangles once per triple; train a 2-bottleneck autoencoder; compare held-out error to a column-shuffled null (several repeats).
- (3) Rank triples by real/null ratio (small = strong candidate).

The autoencoder is purely a screening tool: it flags *whether* a triple is suspiciously dependent, not what the relation is. The `verify_euler_relation` module runs this on (R, r, d) , confirming it flags Euler's relation with no prior knowledge of the formula.